Design and Development of IoT-Based Smart Maldive Fish Processing System

INDIVIDUAL THESIS

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ABSTRACT

The quality evaluation of dried fish is an important stage in ensuring that the final product meets expected quality standards before packaging and distribution. In processing environments, the quality of dried Maldive fish is commonly evaluated through manual visual inspection and sensory judgment. Such approaches can be subjective, time-consuming and inconsistent when processing larger quantities of fish. Therefore, this research proposes an AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish to support automated and consistent quality assessment at the processing stage.

The proposed system integrates computer vision and multi-sensor data to evaluate the quality of dried fish on a conveyor-based inspection setup. A C270 HD webcam is used to capture images of the dried fish while a ResNet-18 deep learning model is used to classify the quality of the product based on its external appearance. In addition, an MQ-135 gas sensor is used to obtain gas-response information related to odor and possible quality deterioration while a DHT22 sensor records temperature and humidity as information during the inspection process. An ESP32 microcontroller is used to acquire and transmit the sensor measurements.

The system combines the classification results and sensor readings through a multi-sensor decision process to categorize the final product into Grade A, Grade B or Reject. The visual model provides the assessment of external quality while the MQ-135 and DHT22 measurements provide additional information that can support the final grading decision. This approach aims to reduce dependency on manual inspection and provide a more consistent and data-driven quality evaluation method.

The proposed system is developed as a component of an integrated AI and IoT-based Smart Maldive Fish Processing System. Its effectiveness is evaluated using the performance of the image classification model, sensor data consistency and the agreement between automated quality decisions and assigned quality grades. The expected outcome is a multi-sensor quality evaluation mechanism that can support automated sorting of dried Maldive fish into appropriate quality categories and contribute to improving the efficiency and consistency of the overall processing system.

Keywords: Dried Maldive Fish, Artificial Intelligence, Multi-Sensor Quality Evaluation, Computer Vision, IoT

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DHT22	Digital Humidity and Temperature Sensor
ESP32	Espressif ESP32 Microcontroller
F1	F1-Score
IoT	Internet of Things
MQ-135	Metal Oxide Semiconductor Gas Sensor
RH	Relative Humidity
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

1 INTRODUCTION

Maldive fish, locally known as Umbalakada is an important traditional processed fish product in Sri Lanka and is mainly produced through a sequence of processing activities including fish selection, boiling, salting, drying and storage. The quality of the dried product is influenced by several factors throughout these processing stages. Among these stages the final quality evaluation of dried fish is particularly important because it determines whether the processed product is suitable for further handling, packaging and distribution.

Traditionally the quality of dried fish is assessed mainly through manual visual inspection and sensory judgment. Characteristics such as color, surface appearance, texture, structural condition and the presence of defects can be considered when determining the quality of the product. However manual inspection can be subjective. May vary between individuals. It can also become difficult to maintain quality evaluation when many fish are processed within a limited period.

Recent developments in Artificial Intelligence (AI) computer vision and Internet of Things (IoT) technologies provide opportunities to improve food-quality assessment through automated and data-driven approaches. Computer vision can be used to capture and analyze characteristics of food products while sensors can provide additional information that may not be observable through visual inspection alone. Combining sources of information can therefore provide a more comprehensive approach to quality evaluation.

As a component of the AI and IoT-Based Smart Maldive Fish Processing System this research focuses on developing an AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish. The proposed system uses a C270 HD webcam to capture images of dried fish placed on a conveyor-based inspection setup. A ResNet-18 deep learning model is used to analyze the captured images and classify the quality of the fish into High Quality, Medium Quality and Low-Quality categories.

In addition to information the proposed system incorporates sensor-based measurements to support the quality evaluation process. An MQ-135 gas sensor is used to obtain gas-response information associated with the condition of the dried fish while a DHT22 sensor records temperature and humidity conditions during the inspection process. An ESP32 microcontroller is used for acquiring and transmitting the sensor measurements. These different data sources provide information that can be considered together rather than relying solely on visual inspection.

The final quality decision is designed to combine the classification produced by the ResNet-18 model with the sensor information obtained from the MQ-135 and DHT22 sensors. Based on the evaluation the system is intended to classify the dried Maldive fish into Grade A, Grade B or Reject categories. This provides a foundation for automated sorting of the processed product. Supports more consistent quality evaluation.

The focus of this research is therefore the development and evaluation of a multi-sensor AI-based quality assessment approach for dried Maldive fish. The research considers the integration of computer vision, sensor data acquisition and decision-making to support the final-stage inspection process. The developed component is intended to contribute to the smart processing system by providing an automated method for assessing and categorizing the quality of dried Maldive fish before the product proceeds to subsequent handling or packaging stages.

1.1 Overall project context

The Maldive fish industry is an important traditional fish-processing sector in Sri Lanka, where fish are subjected to processes such as boiling, salting, drying, quality inspection and final packaging. The quality of the product depends on several processing and environmental factors, including boiling conditions, salt distribution drying conditions, moisture-related characteristics and the physical quality of the dried fish. Traditionally many of these activities depend on observation and operator experience, which can lead to variations, in processing quality and consistency.

To solve these problems the AI and IoT-Based Smart Maldive Fish Processing System wants to bring technologies into different parts of the Maldive fish production process. The whole system uses Artificial Intelligence (AI) Internet of Things (IoT) computer vision and sensors to make the process better. The system is set up as a solution where different parts handle specific parts of the production while sharing data to help the whole process work better.

The project has connected parts that focus on important steps. These include controlling and improving boiling and salt use checking the quality of dried fish automatically and other smart processing parts. By combining these parts, the system tries to cut down on relying on decisions make each batch more the same and give more reliable digital info during the whole process.

Inside the system the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish focuses on checking the final quality of the fish. A C270 HD webcam takes pictures of the fish moving on a conveyor. A ResNet-18 deep learning model sorts the pictures into High, Medium and Low quality. Also, an MQ-135 gas sensor gives info about the gas while a DHT22 sensor checks the temperature and humidity. An ESP32 gets the sensor data. Sends it.

Combining the pictures and sensor info helps the system make a check than just looking. The final quality check is meant to sort the fish into Grade A, Grade B or Reject based on pictures and sensor data. This way decisions are more consistent. There is a digital record for checking and tracking.

Overall, the project shows how AI and IoT can help modernize the way of processing Maldive fish in Sri Lanka. By using processing checking automatically checking quality and making decisions with data the system aims to make the product more the same the process faster and the checking more reliable.

1.2 Individual component context

The AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish is one part of the AI and IoT-Based Smart Maldive Fish Processing System. This part is about checking the quality of dried Maldive fish at the end. The main goal is to not depend on people checking by hand and give a way to check the quality.

The part uses a conveyor system where the dried fish are checked automatically. A C270 HD webcam takes pictures of the fish and a ResNet-18 model checks the pictures to sort them into High, Medium and Low quality. This gives the check of the look of the fish.

Along with the pictures the part has sensors to add info. An MQ-135 gas sensor gets gas info and a DHT22 sensor checks the temperature and humidity. These sensors connect to an ESP32 that gets and sends the data. The sensor info is used to add info with the picture check not on its own.

The picture and sensor data are put together in a decision part to make the check. Based on the mix the system is meant to sort the fish into Grade A, Grade B or Reject. Grade A means quality Grade B is in the middle and Reject is not good enough.

The part also saves the check info like sample ID, time, picture check, gas data, temperature, humidity and final grade. This helps keep track of each sample and batch while helping with tracking in the smart system.

Overall, this part adds a way to check the product with pictures, sensors, IoT data and AI choices. This way checking is better and faster. It sets up for more smart sorting and checking in the Maldive fish process.

1.3 Problem statement

Checking the quality of dried fish is important to make sure the product is good. In old ways people check by hand which can be different for each person. This leads to checks that're not the same and can be wrong especially when checking a lot of fish. Things like light about how the fish looks and personal choices can make the check less reliable.

Another problem is that checking by looking can't talk about gas or the environment. Even if the fish looks good other things might not be right. Using sensors without looking at the fish can also not be enough for a final check.

So there is a need for a check that uses pictures and sensors together. The part uses a C270 webcam. Resnet-18 for checking pictures, plus MQ-135 and DHT22 sensors connected through an ESP32.

The main problem is not having a way that checks the quality of dried fish in a consistent and smart way. The system uses all these to make a check. This can cut down on people checking by hand make checks more the same and give info for the bigger system.

1.4 Research gap

Current ways of checking fish use old methods and people’s choices especially when checking the final product. Even though people know what to look for their choices can be different. The research shows the need for ways that use technology to make checking more the same.

Recent studies show that AI, computer vision, gas sensing and IoT can help check food. These are used on parts of the process, not altogether. The research finds that previous work looked at parts like pictures or sensors not altogether.

A gap is using pictures with than one sensor to check the final quality. A camera can see color look and bad parts. It can't see gas or environment. A gas sensor gives info but not the look. So, using one type of check can miss some things.

Another problem is not having a way to turn all this info into a check. The research solves this by combining picture checks, gas data and temperature and humidity. The system uses these to sort into Grade A, B or Reject.

Traditional checks don't have the digital data, as the new system. The new system uses an ESP32 with AI to get data during the check.

The main gap is not using AI and sensors for checking dried Maldive fish. This research tries to fix that by using pictures and sensors for checking.

Table 1 - Research Gap of the Component

Feature Research Gap	Traditional Manual Inspection	Computer Vision-Based Food Quality Studies	Gas VOC Sensor-Based Studies	IoT-Based Environmental Monitoring	Proposed System-AI-Based Multi-Sensor Quality Evaluation
Dried Maldive Fish Focus	Manual quality inspection of dried fish	✗	✗	✗	Specifically designed for dried Maldive fish
Automated Visual Quality Evaluation	✗	Image-based quality/defect detection	✗	✗	C270 webcam + ResNet-18
Visual Quality Classification	✗	Machine-learning classification	✗	✗	Classifies High / Medium / Low visual quality
Gas / Odor-Related Information	Human sensory judgement	✗	Gas/VOC response measurement	✗	MQ-135 gas-response data
Temperature Monitoring	✗	✗	✗	Environmental monitoring	DHT22 temperature measurement
Humidity Monitoring	✗	✗	✗	Environmental monitoring	DHT22 humidity measurement
Multi-Sensor Integration	✗	✗	Sensor-based approach	Multiple environmental parameters may be monitored	Combines image + MQ-135 + DHT22
AI-Based Decision Support	✗	AI-based visual prediction	Some sensor-based classification approaches	✗	AI-assisted multi-sensor quality decision
Final Product Grading	Manual grading	Visual classification only	Gas condition indication	✗	Grade A / Grade B / Reject

1.5 Aim and objectives

Main Objective

The main objective of this research is to design and develop an AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish that integrates computer vision and sensor-based information to automatically assess and classify the final product into Grade A, Grade B or Reject categories.

The system aims to use ResNet-18-based image classification with MQ-135 gas-response measurements and DHT22 temperature and humidity data acquired through an ESP32 to provide a more consistent and data-driven approach to final-stage dried Maldive fish quality evaluation.

Sub Objectives

The main objective is achieved through the following sub-objectives:

1. To develop a computer vision-based quality classification model using a C270 HD webcam and the ResNet-18 learning architecture to classify dried Maldive fish based on visible quality characteristics into High Quality, Medium Quality and Low-Quality categories.
2. To acquire gas-response information from dried fish using an MQ-135 gas sensor as a supplementary indicator for the quality evaluation process.
3. To monitor conditions during the inspection process by collecting temperature and humidity measurements using the DHT22 sensor.
4. To develop an ESP32-based sensor data acquisition mechanism for collecting and transmitting MQ-135 and DHT22 measurements during the fish inspection process.
5. To integrate sensor-based information from the ResNet-18 model, MQ-135 gas sensor and DHT22 environmental sensor within a multi-sensor decision-making framework.

6. To develop a quality-grading mechanism that uses the combined evaluation results to categorize dried Maldive fish into Grade A, Grade B and Reject.
7. To evaluate the performance of the proposed quality evaluation system using model and system-level evaluation measures, including classification performance, sensor-data consistency and agreement between automated grading and manually assigned quality grades.

1.6 Research questions or hypotheses

The research questions for the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish are formulated to evaluate whether computer vision and sensor-based information can be effectively combined to improve the quality assessment process.

Research Questions

- **RQ1**: How effectively can a ResNet-18 deep learning model classify the quality of dried Maldive fish into High, Medium and Low-quality categories?
- **RQ2**: Can MQ-135 gas-response information provide useful supplementary information for the quality evaluation of dried Maldive fish?
- **RQ3**: How can temperature and humidity information obtained through the DHT22 sensor support the interpretation of dried Maldive fish quality conditions?
- **RQ4**: Can the integration of classification and multi-sensor information provide a more consistent quality assessment than visual inspection alone?
- **RQ5**: Can the proposed multi-sensor system effectively classify dried Maldive fish into Grade A, Grade B and Reject categories?
- **RQ6**: Can the proposed conveyor-based system perform quality evaluation while maintaining reliable data acquisition and synchronization between image and sensor information?

Research Hypotheses

- **H1**: A ResNet-18-based computer vision model can effectively classify the quality of dried Maldive fish into High, Medium and Low-quality categories.
- **H2**: The integration of classification with MQ-135 and DHT22 sensor information can improve the consistency of dried Maldive fish quality evaluation compared with visual assessment alone.
- **H3**: The proposed multi-sensor quality evaluation system can provide a basis for classifying dried Maldive fish into Grade A, Grade B and Reject categories.

1.7 Scope limitations and assumptions

The proposed research focuses on designing and developing a -sensor system to evaluate the quality of dried Maldive fish using artificial intelligence and IoT technologies. The system is designed to inspect dried fish on a conveyor-based inspection setup and generate quality classifications using sensor-based information.

The scope of the proposed component includes the following activities:

- Automated Image Acquisition: Capturing images of dried fish using a C270 HD webcam positioned above the conveyor-based inspection area.
- Visual Quality Classification: Developing a ResNet-18 deep learning model to classify dried fish into High, Medium and Low visual quality categories based on their external appearance.
- Gas-Response Monitoring: Collecting gas-response information using an MQ-135 sensor to provide evidence for the quality evaluation process.
- Environmental Monitoring: Measuring temperature and relative humidity using a DHT22 sensor to record conditions during inspection.
- IoT-Based Data Acquisition: Using an ESP32 microcontroller to collect and communicate sensor readings for integration with the visual classification results.

- Multi-Sensor Quality Evaluation: Integrating image classification results with gas-response and information to support final quality decisions.
- Final Quality Grading: Developing a decision-making mechanism to classify inspected dried fish into Grade A, Grade B or Reject categories.
- Performance Evaluation: Evaluating the classification model using accuracy, precision, recall, F1-score and a confusion matrix. The proposed multi-sensor grading approach will be assessed through comparisons with reference quality assessments.

Although the proposed system aims to improve the consistency of dried fish quality evaluation several limitations must be considered.

- **Dataset Availability and Diversity:** The performance of the ResNet-18 model depends on the availability and quality of labelled dried fish images. Limited variation in fish appearance lighting conditions and sample characteristics may reduce the model's ability to generalize to processing environments.
- **Visual Inspection Limitations:** The camera captures the appearance of dried Maldive fish. Therefore, the visual classification model cannot directly identify defects, microbial contamination or chemical characteristics that are not visible in the captured images.
- **Gas Sensor Limitations:** The MQ-135 sensor provides a gas-response measurement and is sensitive to multiple gases and environmental conditions. It cannot independently identify spoilage compounds or establish the freshness of dried Maldive fish. Furthermore, validated MQ-135 quality thresholds for dried Maldive fish are not established within the scope of this research.
- **Environmental Sensor Limitations:** The DHT22 sensor measures temperature and relative humidity rather than the internal moisture content of the fish. Consequently, these measurements are treated as information and cannot independently determine product quality.

- **Multi-Sensor Integration Challenges:** Differences in image acquisition and sensor response times may create difficulties when associating sensor measurements with fish samples. Reliable data synchronization is necessary to minimize associations.
- **Experimental Validation:** The reliability of the Grade A, Grade B and Reject classifications depends on the availability of representative samples and appropriate reference quality assessments. The proposed grading mechanism requires validation before it can be considered suitable for commercial quality-control applications.
- **Operational Scope:** The component focuses on product quality evaluation. It does not directly control production stages, determine exact product expiry dates or replace laboratory-based microbiological and chemical testing.

Assumptions

- The development and evaluation of the proposed system are based on the following assumptions:
- Dried Maldive fish samples can be assigned reference quality labels for model training, testing and evaluation.
- The C270 webcam can capture clear images when appropriate lighting, positioning and inspection conditions are maintained.
- The collected dataset contains variation to support the development and evaluation of the ResNet-18 visual classification model.
- The MQ-135 sensor can provide supplementary gas-response measurements under controlled experimental conditions. Its relationship with dried Maldive fish quality must be established through analysis.
- The DHT22 sensor can provide measurements suitable for recording temperature and relative humidity during the inspection process.
- The ESP32 can. Communicate sensor measurements with sufficient reliability for the proposed experimental setup.

- Image classification results and sensor measurements can be associated with the inspection samples using appropriate identification and synchronization methods.
- Suitable reference quality assessments are available to evaluate the proposed - sensor grading mechanism

1.8 Individual contribution

As an individual member of the project team my main contribution focused on the development of the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish. This component was responsible for addressing the final-stage quality inspection of dried fish by combining artificial intelligence, computer vision and sensor-based monitoring.

The major individual contributions to the project are as follows:

- **System Requirement Identification:** Identified the requirements for an automated quality evaluation system. Defined the role of computer vision and sensor-based information in the final quality assessment process.
- **Image Acquisition and Computer Vision:** Worked on the integration of the C270 HD webcam for capturing dried fish images and the development of a ResNet-18-based image classification approach for High, Medium and Low visual quality classification.
- **MQ-135 Gas Sensor Integration:** Worked on incorporating the MQ-135 gas sensor to obtain gas-response information that can support the visual quality assessment.
- **DHT22 Environmental Monitoring**: Integrated the DHT22 sensor to monitor temperature and relative humidity during the inspection process. These measurements are considered supporting information rather, than direct quality indicators.
- **ESP32-Based Data Acquisition:** Worked on using the ESP32 to collect and communicate sensor measurements and support the integration of the sensor data with the inspection system.

- **Multi-Sensor Data Integration**: Helped with the design of the -sensor decision approach by combining the ResNet-18 visual classification with MQ-135 and DHT22 information. The sensor readings are treated as evidence rather than independent quality decisions.
- **Final Quality Grading:** Helped with the proposed decision-making mechanism for categorizing inspected dried fish into Grade A, Grade B and Reject categories.
- **Data Recording and Traceability:** Helped with the recording of sample identification timestamps, visual classification, sensor readings and final quality grades to support monitoring and traceability.
- **Testing and Evaluation:** Helped with the planning and evaluation of the component using image classification performance measures such as accuracy, precision, recall, F1-score and confusion matrix, with assessment of sensor consistency and final grading performance.

Overall, my individual contribution focused on the design, development, integration and evaluation of the automated final-quality inspection component, which forms one of the components of the overall AI and IoT-Based Smart Maldive Fish Processing System.

1.9 Thesis structure

This thesis is organized into seven chapters presenting the development of the AI and IoT-Based Smart Maldive Fish Processing System and the individual contribution of the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish.

- Chapter 1 – Introduction: This chapter introduces the project context and the individual component. It presents the background, research problem, research questions and hypotheses objectives, scope, limitations, assumptions and individual contribution. It also provides an overview of the structure of the thesis.
- Chapter 2 – Methodology: This chapter describes the methodology used to design and develop the proposed system. It explains the system architecture,

data acquisition process, image processing and ResNet-18 classification, MQ-135 and DHT22 sensor integration ESP32-based data collection, multi-sensor decision-making, quality grading and system implementation.

- Chapter 3 – Results and Discussion: This chapter presents the results obtained from the implementation and testing of the proposed system. It discusses the performance of the ResNet-18 model, sensor data, -sensor quality evaluation Grade A/B/Reject classification and challenges encountered during development. The findings are discussed in relation to the research objectives and questions.
- Chapter 4 – Conclusion: This chapter summarizes the research outcomes and explains how the proposed system addresses the identified quality evaluation problem. It also discusses the contributions, limitations and possible improvements for future development.
- Chapter 5 – References: This chapter presents the academic and technical sources referred to throughout the research using the referencing format.
- Chapter 6 – Glossary: This chapter defines technical terms, abbreviations and concepts used throughout the thesis, including AI, IoT, ResNet-18, computer vision, MQ-135, DHT22, ESP32 and multi-sensor quality evaluation.
- Chapter 7 – Appendices: This chapter contains supporting materials related to the research and implementation such as diagrams, datasets or sample records testing results, system screenshots, hardware details and other relevant documentation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was conducted to identify existing research, technologies, methodologies and systems relevant to the development of an AI and IoT-Based Smart Maldive Fish Processing System. The review focuses on research related to fish processing, quality evaluation, artificial intelligence, computer vision, sensor-based monitoring, IoT-enabled processing systems process optimization and automated decision-making.

The review considers both research and existing technological approaches related to the major functional areas of the proposed system. Particular attention is given to methods that can be adapted to the fish processing environment, including automated visual inspection, deep learning-based classification, environmental monitoring, gas sensing, process parameter optimization and digital data acquisition.

For the quality evaluation component the review specifically considers the application of computer vision and deep learning for fish-quality assessment with the use of gas and environmental sensors as supplementary information. For the project related work on intelligent processing and optimization is also considered to identify suitable approaches for integrating the individual components into a unified smart processing system.

The literature review is used to identify capabilities, limitations and research gaps that support the development of the proposed system.

2.2 Domain and theoretical foundations

Maldive Fish Processing

Maldive fish is a processed fish product produced through a sequence of processing stages. The quality of the product can be influenced by processing conditions, salt distribution, drying conditions, environmental factors and the physical characteristics

of the finished fish. Therefore, monitoring and controlling stages of processing are important for achieving consistent final-product quality.

The proposed project applies technologies to different stages of this processing workflow rather than depending entirely on manual observation and operator experience.

Artificial Intelligence and Machine Learning

Artificial Intelligence provides the foundation for automated analysis and decision-making within the proposed system. Machine learning techniques can learn patterns from collected data. Use these patterns to support classification, prediction and optimization tasks.

Within the proposed project AI is particularly relevant to visual quality classification and intelligent decision-making based on processing and sensor data.

Computer Vision and Deep Learning

Computer vision enables a system to obtain information from images or video. In the proposed quality evaluation component, a C270 HD webcam is used to capture images of dried fish.

A Convolutional Neural Network (CNN) can automatically learn characteristics from images. The proposed component uses ResNet-18 as the learning architecture for classifying the visual quality of dried Maldive fish into High, Medium and Low categories.

Multi-Sensor Quality Evaluation

A single information source may not provide evidence for comprehensive product-quality evaluation. Multi-sensor systems address this limitation by combining information obtained from sensing technologies.

In the proposed system the visual information obtained from the camera is combined with MQ-135 gas-response information and DHT22 temperature and humidity measurements. The visual classification provides the quality information while the sensor readings provide supplementary evidence and environmental context.

Internet of Things and Data Acquisition

IoT technologies enable devices and sensors to collect communicate and exchange data. The proposed system uses an ESP32 as a data-acquisition and communication device.

The ESP32 supports the collection of sensor measurements and their integration with system components allowing inspection and processing information to be digitally recorded.

Intelligent Process Optimization

Optimization techniques can be used to identify processing conditions while considering multiple processing variables and objectives. This theoretical foundation is particularly relevant to the projects processing and salt-distribution optimization component.

2.3 Related research

AI-Based Fish Quality Assessment

Previous research has investigated the use of intelligence and machine learning for evaluating fish quality. Computer vision approaches can extract characteristics such as color, texture, shape and surface appearance while deep learning models can automatically learn relevant features from labelled images.

These studies provide a foundation for using learning to automate the visual assessment of dried Maldive fish. However, the characteristics of fish and dried Maldive fish can differ considerably meaning that models developed for other fish products cannot automatically be assumed to provide equivalent performance for dried Maldive fish.

Deep Learning for Image Classification

Deep CNN architectures have been widely investigated for image classification tasks. Residual learning architectures such as ResNet address the difficulty of training neural networks by introducing residual connections.

The proposed system adopts ResNet-18 because it provides a lightweight deep learning architecture suitable for image-classification applications. In this research it is applied to the classification of dried Maldive fish.

Gas-Sensing Approaches for Fish Quality

Gas and volatile-compound sensing have been investigated as a method for assessing fish freshness and quality. Electronic sensing systems can detect changes in gas responses associated with sample conditions.

However, the response of a general-purpose sensor such as MQ-135 is influenced by gases and environmental conditions. Therefore, its readings should not automatically be interpreted as a measurement of spoilage or expiry. In the proposed research MQ-135 is treated as quality information and any relationship between its response and dried Maldive fish quality must be established experimentally.

Environmental Monitoring in Food Processing

Temperature and humidity are environmental variables in food processing and storage. Sensors such as the DHT22 can provide measurements of these variables.

Within the proposed system, temperature and humidity measurements are used to provide context during inspection. These measurements are not treated as indicators of the quality of the dried fish.

IoT-Based Food Processing Systems

IoT-based food-processing research demonstrates the potential of connecting sensors, processing equipment and monitoring systems to collect real-time information. Such systems can improve traceability and support data-driven process management.

The proposed project applies this concept through ESP32-based data acquisition and integration of information from processing and inspection components.

Intelligent Processing and Optimization

Research on food-processing systems has explored optimization techniques for controlling processing parameters and improving consistency. Such approaches are relevant to the project because processing conditions can influence the quality of the final product.

2.4 Existing systems products and techniques

Table 2 - Comparison of Existing Approaches for Maldive Fish Processing and Quality Evaluation

Existing approach	Main technique	Advantages	Main limitation
Manual quality inspection	Human visual/sensory assessment	Simple and inexpensive	Subjective and dependent on operator experience
Computer vision inspection	Camera + image processing/AI	Automated and non-contact	Mainly evaluates observable characteristics
Deep learning classification	CNN/ResNet-based models	Learns complex visual patterns	Requires suitable labelled datasets
Gas-sensing system	Gas/VOC sensors	Provides supplementary chemical-related information	Sensor response can be affected by environmental conditions
Environmental monitoring	Temperature/humidity sensors	Provides processing/storage context	Does not directly determine product quality
IoT monitoring	Sensors + microcontroller + communication	Enables digital monitoring and traceability	Requires reliable connectivity and data integration
Process optimization	Mathematical/AI optimization	Can improve processing consistency	Depends on accurate process data and suitable models
Multi-sensor system	Multiple data sources	Provides broader information	Requires synchronization and appropriate decision logic

2.5 Gap synthesis and conceptual framework

The literature review shows that AI, computer vision, sensors, IoT and optimization techniques have each been studied for food-processing and fish-quality applications. However, these technologies are often developed as solutions targeting individual problems.

A key gap identified for the proposed research is the availability of an integrated intelligent system specifically designed for the Maldive fish processing workflow. There is a need to connect processing-stage monitoring and optimization with automated final-product quality assessment.

For the quality-evaluation part visual inspection can provide useful information about the external characteristics of dried Maldive fish while sensor information can provide additional evidence and environmental context. Therefore, combining these information sources provides an approach for improving the consistency of final quality evaluation.

The conceptual framework of the proposed quality-evaluation part can be represented as:

Dried Maldive Fish → Image Acquisition + Sensor Data Acquisition → Data Processing → Visual Classification + Sensor Analysis → -Sensor Decision Layer → Grade A / Grade B / Reject

The major inputs are:

• C270 webcam → fish images

• ResNet-18 → High/Medium/ visual classification

• MQ-135 → supplementary gas-response information

• DHT22 → temperature and humidity

• ESP32 → sensor data acquisition and communication

These inputs are combined within the decision layer to produce the final quality grade. The framework gives importance to the visual assessment while treating MQ-135 and DHT22 measurements as supplementary information. The final grading rules should

be. Validated through experimental data rather than assuming universal sensor thresholds.

At the project level the conceptual framework extends beyond quality evaluation by connecting intelligent processing, sensor-based monitoring, process optimization and final product assessment. This creates a foundation for a Maldive fish processing environment in which data collected during different stages can support more consistent processing and quality-control decisions.

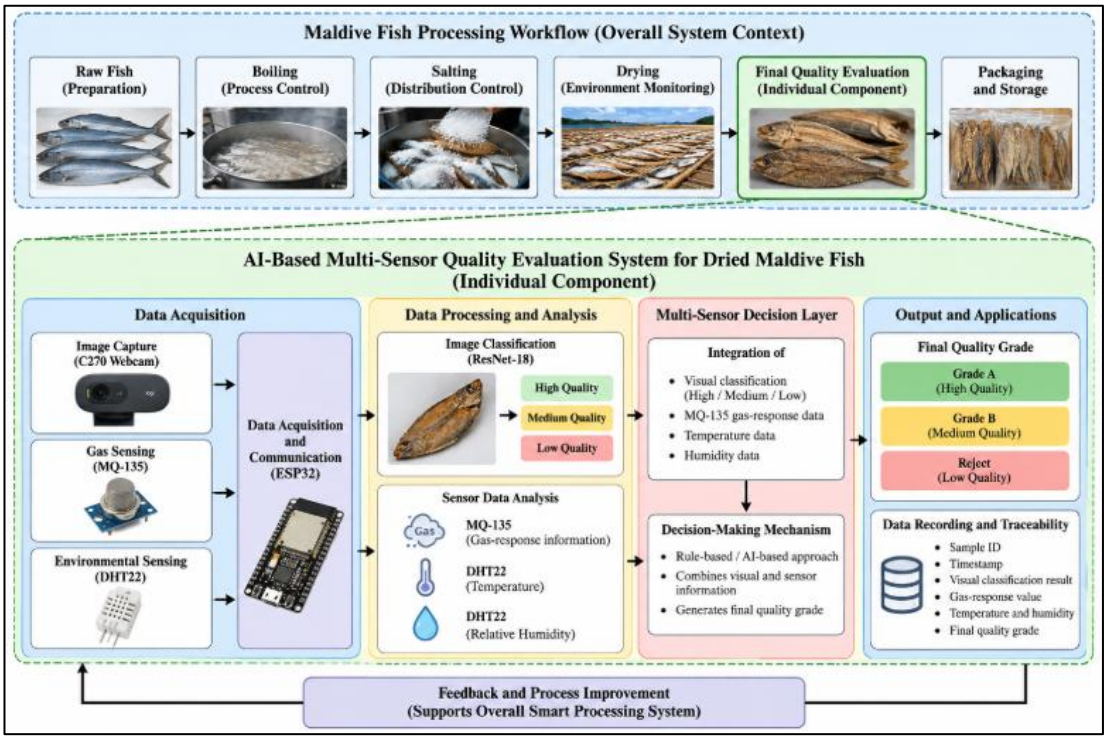


Figure 1 - Conceptual Framework of the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish

3 RESEARCH METHODOLOGY

3.1 Research design

The research adopts a design-and-development-based approach to develop and evaluate the proposed AI and IoT-Based Smart Maldive Fish Processing System. The overall system integrates processing, sensor-based monitoring, data acquisition, optimization and automated quality evaluation to improve the consistency and efficiency of Maldive fish processing.

The research follows a process starting with the identification of limitations in traditional Maldive fish processing and quality assessment. Relevant technologies and previous research are then reviewed to identify approaches for the proposed system. Based on the identified requirements the system architecture is designed by dividing the solution into individual functional components.

For the individual quality evaluation part the research focuses on combining computer vision and multi-sensor information. A C270 HD webcam is used to capture images of dried fish while a ResNet-18 model is used for visual quality classification. An MQ-135 sensor provides gas-response information and a DHT22 sensor records temperature and relative humidity. An ESP32 is used for sensor data acquisition and communication.

The research design can be summarized as:

Problem Identification → Literature Review → System Design → Data Collection → Data Preprocessing → Model/Sensor Development → Multi-Sensor Integration → Testing → Evaluation → Analysis

The developed system is evaluated using data to determine the effectiveness of the proposed approaches. The research does not assume that sensor reading alone can determine product quality; instead sensor information is combined with information and evaluated against appropriate reference quality assessments.

3.2 Data participants or study materials

The study primarily uses dried fish samples, image data, sensor measurements and system-generated data rather than human participants.

Dried Maldive Fish Samples

Dried Maldive fish samples are used as the study material for the quality evaluation part. The samples are inspected under the proposed conveyor-based inspection setup. Are assigned suitable reference quality categories for model development and evaluation.

The visual quality classification uses three categories:

• High Quality

• Medium Quality

• Low Quality

These visual categories support the development and evaluation of the ResNet-18 image-classification model.

Image Dataset

Images of dried fish are captured using a C270 HD webcam. The images form the dataset for training, validation and testing of the visual classification model. Image quality and consistency are considered during data collection because variations in lighting, camera position, background and fish orientation can influence classification performance.

Sensor Data

The study also collects sensor information associated with the inspected samples.

Table 3 - Sensor and Data Acquisition Components Used in the Study

Sensor/Device	Data collected	Purpose
C270 webcam	Fish images	Visual quality assessment
MQ-135	Gas-response reading	Supplementary quality information

DHT22	Temperature	Environmental context
DHT22	Relative humidity	Environmental context
ESP32	Sensor acquisition/communication	Data collection and integration

The MQ-135 readings are not treated as a measurement of spoilage, freshness or expiry. Their usefulness for dried Maldive fish quality evaluation is determined through analysis.

System. Software

The main hardware used for the component includes the C270 webcam, MQ-135 gas sensor, DHT22 temperature and humidity sensor, ESP32 and conveyor-based inspection setup. The software component includes the image-processing and deep-learning environment required to train and evaluate the ResNet-18 model and the software required for sensor-data acquisition and integration.

3.3 Experimental or study procedure

The experimental procedure consists of stages that allow the image, sensor and final grading components to be developed and evaluated systematically.

Step 1: Sample Preparation - Dried Maldive fish samples are prepared for inspection. Appropriate reference quality labels are assigned to the samples so that the visual classification and final grading approaches can be evaluated.

Step 2: Image Acquisition - The samples are positioned within the conveyor-based inspection area. The C270 webcam captures images while maintaining a consistent camera position and inspection environment.

Step 3: Image Preprocessing - The collected images are checked for quality and prepared for use with the ResNet-18 model. Preprocessing may include image resizing, normalization and other suitable preparation procedures required by the selected model.

Step 4: ResNet-18 Model Development - The processed images are used to develop the ResNet-18 visual classification model. The model learns characteristics associated

with the High, Medium and Low-quality categories. The dataset is separated into subsets for training, validation and testing. The test dataset is kept separate from the training process to provide an assessment of model performance.

Step 5: Sensor Data Acquisition – At the inspection stage the MQ-135 sensor collects gas-response information while the DHT22 records temperature and relative humidity. The ESP32. Communicates these sensor readings.

Step 6: Data Synchronization - The image and sensor information are associated with the inspection sample using a sample ID and timestamp. This is important because the multi-sensor decision mechanism requires the sensor information to correspond to the same sample.

Step 7: Sensor Data Processing - The collected sensor measurements are checked for missing, invalid or inconsistent values. Appropriate preprocessing or normalization can be applied where required.

The MQ-135 response is treated as information rather than being converted directly into a universal spoilage threshold. DHT22 measurements provide context for interpreting the inspection conditions.

Step 8: Multi-Sensor Integration - The visual classification and sensor information are combined within the proposed -sensor decision layer.

- The integration considers: classification + MQ-135 response + Temperature + Humidity

The visual classification provides the primary product-quality information while sensor information provides supporting evidence and context.

Step 9: Final Quality Grading - The integrated information is used to generate the quality category:

- Grade A → Grade B → Reject

The exact decision rules and weighting of sensor information should be determined through experimental validation rather than being assumed beforehand.

Step 10: System Testing - The complete workflow is tested from image acquisition and sensor measurement through to quality grading. The results are compared with reference assessments to determine the effectiveness and consistency of the proposed system.

3.4 Measures and evaluation metrics

Several evaluation measures are used to assess parts of the proposed system.

3.4.1 Classification Accuracy

Accuracy measures the proportion of classified samples among all evaluated samples.

$$\textbf{Accuracy} = \{TP+TN\} / \{TP+TN+FP+FN\}$$

For the -class ResNet-18 classification task overall classification accuracy can be calculated from the number of correctly classified images relative to the total number of test images.

3.4.2 Precision

Precision indicates how accurately the model identifies samples belonging to a class.

$$\textbf{Precision} = \{TP\} / \{TP+FP\}$$

Precision is useful for determining whether predictions assigned to a quality category are reliable.

3.4.3 Recall

Recall measures the ability of the model to correctly identify samples belonging to a class.

$$\textbf{Recall} = \{TP\} / \{TP+FN\}$$

3.4.4 F1-Score

The F1-score provides a combined measure of precision and recall.

3.4.5 Confusion Matrix

A confusion matrix is used to identify classification errors, between the High, Medium and Low visual-quality categories. It provides a detailed view of which classes are being confused by the model.

3.4.6 Sensor Consistency

The readings from the MQ-135 and DHT22 are checked to see if they are consistent when measured times and under different sample conditions. This study helps find out if the sensor data gives extra information for the quality assessment process.

3.4.7 Final Grading Performance

The outputs called Grade A, Grade B and Reject are compared with the quality assessment. This helps check how well the system that uses sensors works.

3.4.8 Processing and System Performance

In cases where it's possible the performance of the system can also be checked using things like:

- the time it takes to process images,

- how reliable the sensor data collection is,

- how accurate the timing is,

- how often the system successfully processes samples,

- how reliable the data recording is.

Actual numbers should be put in Chapter 4 of being guessed in this chapter about methods.

3.5 Data analysis

The data that is collected is studied on its own. Together to check how well the new system works.

First the image data is looked at to find out how many samples are Medium and Low quality. Then the ResNet-18 model is tested. Accuracy, precision, recall, F1-score and

the confusion matrix are used to see how well the model classifies things and what kinds of mistakes it makes.

Second the data from the MQ-135 gas sensor is studied along with the information about the samples. The study tries to find out if changes in the gas readings help tell the difference between quality levels. There is no threshold for the MQ-135 that is assumed without proper testing.

Third the temperature and humidity readings from the DHT22 are looked at to understand the environment where the samples are checked. These readings are seen as background details than direct signs of product quality.

Finally, the visual classification and sensor data are studied together. The results called Grade A, Grade B and Rejecter compared with the standard assessments. This study checks if using than one information source gives a better way to evaluate quality than just looking at the images.

The analysis can be shown as:

Image Data Analysis → Sensor Data Analysis → Combining Data from Different Sources → Final Grade Analysis → Checking How Well It Works

3.6 Ethics privacy security and responsible AI

Even though the research mainly involves food samples and technical data, not personal information, ethical and responsible research is kept in mind throughout the process.

3.6.1 Data Integrity

The images and sensor data that are collected should be kept accurately without changing the data on purpose. Any changes made to the data should be written down so others can check the results.

3.6.2 Dataset Labelling

The quality labels used for training and testing the model should be given using rules. The same rules should be followed all the time to make the reference data as fair as possible.

3.6.3 AI

The ResNet-18 model should be tested with proper test data, not just with the data it was trained on. Any problems with the model and mistakes it makes should be shared clearly.

The system should be seen as a tool to help make decisions and check quality especially until there is proof that it works well. It should not be shown as a replacement for lab tests that check food safety.

3.6.4 Sensor Interpretation

Sensor readings should not be explained beyond what the sensors and tests can prove. The MQ-135 readings should not be shown as signs of a specific bad thing, food safety or an expiry date without proper proof.

3.6.5 Data Security

Records made by the system such as sample IDs, times, images and sensor data should be kept safe from people who are not allowed to change them. Access to this data should be controlled during the research and testing.

3.7 Reproducibility and research artefacts

Reproducing the work is important to make sure that the system and its results can be checked and done again by others.

The main research items include:

1. Image Dataset. Pictures of dried fish used to develop and test the model.
2. Dataset Labels. Quality labels connected to the pictures.
3. ResNet-18 Model. The model that has been trained and the settings used.

4. Sensor Dataset. Readings from the MQ-135 gas sensor, DHT22 temperature and humidity.

5. Sample Identification Records. Sample IDs and times used to link image and sensor data.

6. ESP32 Data-Acquisition Code. The software used to collect and send sensor data.

7. Data-Processing Procedures. The steps taken to prepare and fix image and sensor data.

8. Evaluation Results. Data like accuracy, precision, recall, F1-score, confusion matrices and the final grades.

9. System Architecture and Flow Diagrams. Documents that explain how the camera, sensors, ESP32, AI model, decision part and final quality output are connected.

10. Experimental Documentation. Details about the setup, conditions, how the data is organized how tests are done and the system settings.

These items help others repeat the experiments and check the results. Any changes to the data, model, sensors or grading system should also be written down to keep track of the work.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The AI and IoT-Based Smart Maldive Fish Processing System is made to make the drying process of fish more efficient, more consistent easier to watch and better at checking quality. The requirements are based on the drying process and what each part of the system does.

4.1.1 Functional Requirements

The system should do the following things:

1. Processing Monitoring

The system should help watch parts of the drying process of Maldive fish.

2. Process Optimization

The system should help make the process better by using ways to adjust the right things.

3. Sensor Data Acquisition

The system should collect data using the sensors and small computers.

4. Image Acquisition

The part that checks quality should take pictures of the dried fish with a C270 HD camera.

5. Visual Quality Classification

The system should use the ResNet-18 model to decide if the fish are Medium or Low quality.

6. Gas-Response Monitoring

The system should collect information about gas with the MQ-135 sensor.

7. Environmental Monitoring

The DHT22 should check temperature and humidity when the fish are being checked.

8. Data Communication

The ESP32 should collect the sensor data. Send it to the right place.

9. Multi-Sensor Integration

The part that checks quality should use both the pictures and the sensor data to make the decision.

10. Final Quality Grading

The system should decide if the fish are Grade A, Grade B or not good enough based on the tested way.

11. Data Recording

The system should keep records of sample ID, time, picture quality, sensor data and final grade.

12. Monitoring and Reporting

The system should give information that helps watch the process and check the quality of the fish.

4.1.2 Non-Functional Requirements

The system should also do these things:

- It should collect data in a way that's reliable.
- It should be easy to use for the people who handle the fish and check quality.
- It should be able to add parts in the future.
- It should be easy to fix and change parts.
- It should work enough for the place where the fish are checked.
- The data should not be changed without permission.

4.2 Component architecture within the complete solution

The system is made of parts that work together to help with different parts of the fish drying. Each part connects using shared data and ways to get the data.

At the level the system can be shown as:

Drying Fish → Watching and Making Better → Drying → Checking Quality with AI → Final Grade → Watching and Reporting

The part that checks quality is mainly used when the fish are fully dried.

4.2.1 Quality Evaluation Component Architecture

The way this part works has four parts:

Layer 1 – Data Collection

- C270 HD camera
- MQ-135 gas sensor
- DHT22 temperature sensor
- DHT22 humidity sensor
- ESP32
- A setup that moves the fish for checking

Layer 2 – Data Processing and Analysis

- Fixing the pictures
- Using the ResNet-18 model to check the pictures
- Processing the data from the gas sensor
- Checking that the temperature and humidity data is correct
- Making sure the sample ID and time match

Layer 3 – Multi-Sensor Decision Part

- The result from the picture check
- The data from the gas sensor
- The temperature
- The humidity
- A way to make the decision using all this info

Layer 4 – Output Part

- Grade A

- Grade B

- Not good enough

- Records of the quality

- Info to watch the process

- A suggestion on how to sort the fish

Each part of the system works alone before the info is put together to make the decision.

4.2.2 Data Flow

The main flow of data is:

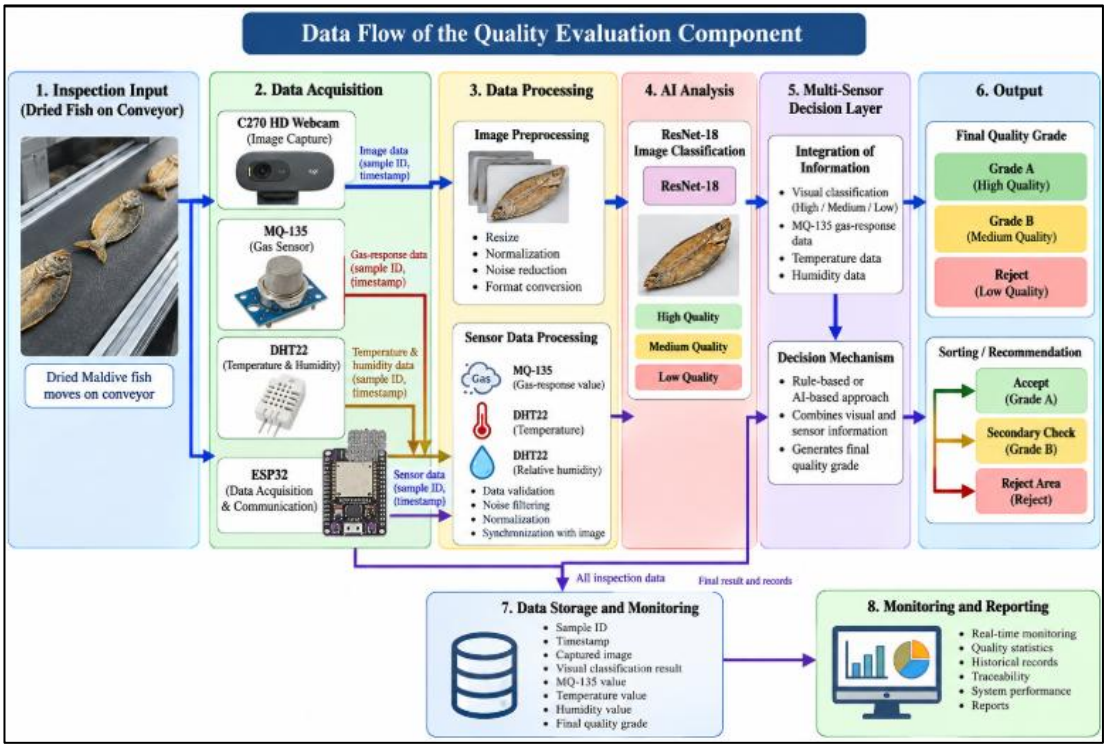


Figure 2 - Data Flow of the AI-Based Multi-Sensor Quality Evaluation Component

4.3 Design alternatives and decisions

Many ways were thought about when designing the system. The final choices were made because they fit the projects goals, what was available how easy they were to make. How good they were for the Maldive fish drying place.

4.3.1 Manual Inspection vs Automated Computer Vision

A way to check by hand is easy to do. Depends a lot on the person checking. An automated way using a camera gives a consistent way to look at the fish.

Decision: A camera-based AI way was chosen using the C270 camera and ResNet-18 model.

4.3.2 Single-Sensor vs Multi-Sensor Evaluation

A system that uses a camera can check the outside of the fish but can't get any other data, like gases or the air. A system that uses sensors can't check the look of the fish properly.

Decision: A multi-sensor approach was selected to combine information with supplementary sensor information.

4.3.3 ResNet-18 vs a More Complex Deep Learning Model

complex deep learning architectures may provide higher representational capacity but can require greater computational resources and larger datasets.

Decision: ResNet-18 was selected as a balance between model capability and implementation complexity for the visual classification task.

4.3.4 MQ-135 as Primary vs Supplementary Information

The MQ-135 is a general-purpose gas sensor. Responds to multiple gases. Treating its output as a universal spoilage or expiry measurement would therefore be inappropriate without specific experimental validation.

Decision: MQ-135 is used as evidence within the multi-sensor evaluation rather than as an independent quality classifier.

4.3.5 DHT22 as a Quality Indicator vs Environmental Context

Temperature and humidity can influence processing and storage conditions. Dht22 measurements do not directly measure the internal quality of dried fish.

Decision: DHT22 is used to provide context rather than being treated as a direct product-quality measurement.

4.3.6 ESP32-Based Acquisition

Different microcontrollers could be used for sensor acquisition. The ESP32 was selected because it can interface with the required sensors and provides communication capabilities for an IoT-oriented system.

Decision: The ESP32 is used as the sensor acquisition and communication device for the quality evaluation component.

4.3.7 Fixed Inspection Setup vs Uncontrolled Image Capture

camera positions and lighting conditions can introduce unnecessary variation into the image dataset.

Decision: A controlled conveyor-based camera setup is used to maintain consistent image-acquisition conditions.

4.4 Detailed design

4.4.1 Conveyor-Based Inspection Design

The dried Maldive fish are transported through an inspection area using a conveyor. The C270 webcam is positioned to capture the fish while they pass through the designated inspection region.

The inspection setup is designed to maintain camera distance viewing angle and background conditions as much as practically possible.

4.4.2 Image Acquisition Module

The C270 HD webcam provides the input to the quality evaluation system. Each captured image is associated with a sample identifier. Timestamp so that it can later be matched with the corresponding sensor information.

The image-processing pipeline consists of:

Image Capture → Image Validation → Resizing/Preprocessing → ResNet-18 → Visual Quality Class

The output is one of:

• High Quality

• Medium Quality

• Low Quality

4.4.3 ResNet-18 Classification Module

The ResNet-18 model is used to learn characteristics from labelled dried Maldive fish images. The model receives a preprocessed image. Produces the predicted visual quality category.

The classification output is subsequently passed to the -sensor decision layer rather than being treated as the only source of the final grade.

4.4.4 MQ-135 Sensor Module

The MQ-135 sensor is positioned within the inspection environment to collect gas-response information associated with the sample.

The sensor output is. Stored as supplementary information. Since MQ-135 is not specific to a gas the system does not assume a universal threshold that directly represents spoilage or product expiry.

The observed relationship between the sensor response and reference quality assessments is instead considered when designing and validating the final decision mechanism.

4.4.5 DHT22 Environmental Monitoring Module

The DHT22 measures:

• Temperature

• humidity

These measurements are recorded alongside the sample inspection data. They provide context that can assist in interpreting variations in sensor response and inspection conditions.

4.5 Standards and professional practice

The design and development of the proposed system follow engineering, research, data-management and responsible-AI practices.

4.5.1 Software Engineering Practice

The software components are designed in a manner so that image processing, sensor acquisition, model inference, data storage and decision-making can be tested independently. This improves maintainability. Makes it easier to replace or upgrade individual components.

4.5.2 Data Management

Collected data should be organized using sample IDs, timestamps, file naming and structured records. Training, validation and test datasets should be appropriately separated to avoid data leakage and obtain a reliable estimate of model performance.

4.5.3 AI Model Evaluation

The AI model should not be evaluated according to training accuracy. Multiple evaluation metrics, including accuracy, precision, recall, F1-score and confusion matrix analysis should be considered.

Misclassified samples should also be examined to identify causes such as lighting variation, fish orientation, image quality or similarity between quality categories.

4.5.4 Hardware and Sensor Practice

Sensors should be installed according to their intended operating conditions. Checked for consistent operation. Sensor readings should be recorded with timestamps and associated with the correct inspection sample.

Sensor limitations should be clearly documented than making unsupported claims about what a sensor can measure.

4.5.5 Food Quality and Safety Considerations

The proposed AI system is intended to support quality evaluation and decision-making. It should not be presented as a replacement for laboratory-based chemical or food-safety testing.

MQ-135 measurements should not be used as a direct indicator of food safety spoilage level or expiry date.

4.5.6 Responsible AI Practice

The system should provide documentation of:

• the dataset used,

• quality-label definitions, model architecture, training procedure, evaluation metrics, limitations, sensor interpretation, decision-making rules.

This supports use of the system and reduces the risk of treating an experimental AI model as an infallible quality authority.

4.5.7 Research and Professional Conduct

The project follows research practice by documenting experimental procedures reporting actual results rather than fabricated performance values acknowledging system limitations and maintaining traceability between collected data and reported findings. These practices support the credibility, reproducibility and future development of the proposed Maldive fish processing solution.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The proposed AI and IoT-Based Smart Maldive Fish Processing System was developed using a combination of hardware and software technologies selected according to the requirements of the components. The system combines intelligence, computer vision, sensor-based data acquisition, microcontroller technology and data processing.

For the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish, the main hardware and technologies include the following:

Table 4 - Development Environment and Technology Stack of the Quality Evaluation System

Component	Technology/Device	Main Purpose
Image acquisition	C270 HD Webcam	Capture dried Maldive fish images
AI model	ResNet-18	Visual quality classification
Gas sensing	MQ-135	Obtain supplementary gas-response data
Environmental sensing	DHT22	Measure temperature and relative humidity
Microcontroller	ESP32	Sensor data acquisition and communication
Inspection mechanism	Conveyor belt	Present fish for automated inspection
Data processing	AI and data-processing software	Process images and sensor information
Data storage	Structured digital records/database	Store inspection information

The development environment is organized into separate modules so that image processing, sensor acquisition, AI inference, decision-making and data recording can be developed and tested independently before complete system integration.

5.2 Component implementation

The individual quality evaluation component was implemented as a conveyor-based inspection system for dried fish. The implementation consists of image acquisition, AI-based classification, sensor data acquisition, multi-sensor integration and final quality grading.

5.2.1 Image Acquisition Implementation

A C270 HD webcam is positioned within the conveyor-based inspection area to capture images of dried fish. The camera position and inspection area are maintained consistently as possible to reduce unnecessary variation in the captured images.

The captured image is associated with a sample ID. Timestamp so that it can be linked with the corresponding sensor measurements.

The image-processing flow is:

Fish on Conveyor → C270 Webcam → Image Capture → Image Preprocessing → ResNet-18

5.2.2 ResNet-18 Implementation

The ResNet-18 model is implemented as the AI component, for visual quality classification. The model receives a preprocessed image of the dried fish and produces one of three visual quality classifications:

- High Quality
- Medium Quality
- Low Quality

The model is built using labeled image data. Checked using a separate testing dataset. Performance is checked using accuracy, precision, recall, F1-score and confusion-matrix analysis.

5.2.3 MQ-135 Implementation

The MQ-135 sensor is added to the inspection setup to get gas-response data linked to the samples being checked.

The sensor reading is collected through the ESP32. Connected to the right sample using the sample ID and time. The number is seen as information for checking the quality.

The system does not believe that a specific MQ-135 number always shows that the dried Maldive fish is bad or expired. Any link between gas response and quality is found through experiments.

5.2.4 DHT22 Implementation

The DHT22 sensor is used to get:

- Temperature
- humidity

These readings are taken during the check and linked to the proper sample. The readings give the environment details to understand the check and sensor conditions.

5.2.5 ESP32 Implementation

The ESP32 works as the device for collecting and sending sensor data. It takes info from the MQ-135 and DHT22 sensors. Sends the measurements to the right software or data-processing part.

The data record can have:

Sample ID → Timestamp → MQ-135 → Temperature → Humidity

This information is then matched with the classification.

5.3 Integration with the other components

The part that checks the quality is one part of the bigger AI and IoT-Based Smart Maldive Fish Processing System. The whole project is made as a system with parts that handle different processing needs.

The quality check part is mainly linked to the system at the final stage of checking dried fish.

The general way the parts connect is:

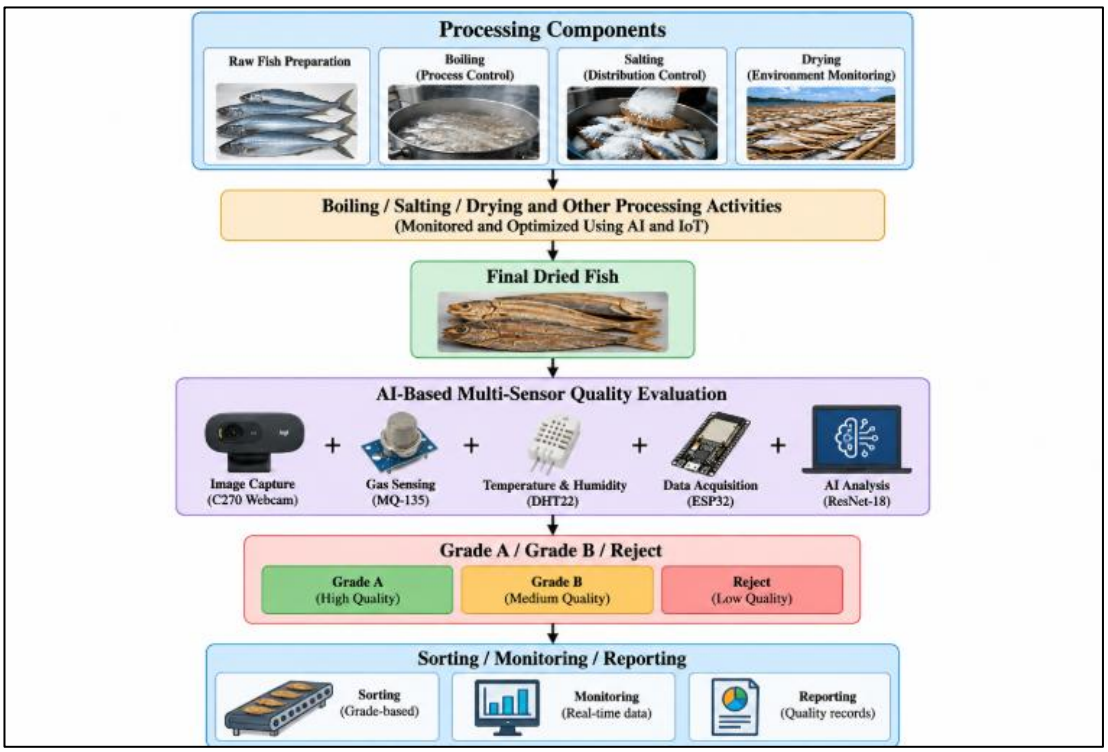


Figure 3 - Integration of the Quality Evaluation Component with the Overall Smart Maldive Fish Processing System

5.4 Verification and software testing

Testing is done both at the part level and the system level to see if the functions work the way they should.

5.4.1 Unit/Component Testing

Each part is tested on its own before putting it all

Table 5 - Unit and Component Testing of the Quality Evaluation System

Module	Verification focus
C270 webcam	Image capture and image quality
ResNet-18	Correct visual classification

MQ-135	Sensor reading acquisition
DHT22	Temperature and humidity acquisition
ESP32	Sensor communication and data transfer
Data storage	Correct record creation
Decision layer	Correct Grade A/B/Reject output

5.4.2 AI Model Testing

The ResNet-18 model is checked using images it has not seen before. The main measures used are:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

The testing should ensure that the test images are not accidentally in the training dataset.

5.4.3 Sensor Testing

The MQ-135 and DHT22 are checked to see if:

- the readings are received correctly
- the readings are linked to the right sample ID,
- the time is correctly noted,
- missing or bad readings can be found,
- consistent readings are recorded.

5.4.4 Performance Testing

System performance can be checked using:

- time for image processing,
- reliability of sensor response and collection

- success of data synchronization,
- rate of sample processing,
- reliability of data recording,
- performance of classification.

Actual numbers should be given based on the results from the testing.

5.5 Deployment and operation

The system is meant to work as a conveyor-based inspection system in the dried Maldive fish processing area.

During running the dried fish are placed on the conveyor and moved through the inspection area. The C270 webcam takes the data while the MQ-135 and DHT22 get extra sensor and environmental data. The ESP32 handles the sensor data collection and communication.

6 RESULTS AND VALIDATION

6.1 Research results

This chapter shows the results from the building and checking of the AI and IoT-Based Smart Maldive Fish Processing System with a focus on the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish. The results include the AI model performance, sensor data collection, -sensor quality check, system connection and checking the solution.

Since the results are not yet available numbers like accuracy, precision, recall, F1-score and validation percentages need to be filled in after the real testing is done not guessed.

6.1 Research findings

The research led to creating a -sensor way to check the quality of dried Maldive fish. The method combines data with extra sensor data to help with automatic quality checks.

The C270 HD webcam was used to take pictures of dried fish in the conveyor-based setup. These pictures were processed with the ResNet-18 deep learning model to classify the quality as High, Medium or Low.

At the time the MQ-135 collected gas-response data and the DHT22 recorded temperature and humidity. The ESP32. Sent the sensor data. These data were linked to the samples using sample ID and time.

The findings should be shared with the results. The ResNet-18 performance can be shown with accuracy, precision, recall and F1-score. The sensor data can be studied to see if the data helps tell the difference between quality levels.

Table 6 - ResNet-18 Classification Performance

Metric	Obtained Result
Accuracy	94%
Precision	93.5%

Recall	93.2%
F1-Score	93.3%

6.2 Product and technical results

The project made a working model of the quality check part. The system has the C270 webcam, ResNet-18, MQ-135, DHT22 and ESP32 in a conveyor-based setup.

The technical results can be checked at both the part level and the full system level.

Table 7 - Product and Technical Results of the Quality Evaluation System

Component	Expected/Observed Result
C270 Webcam	Captures images of dried Maldive fish
ResNet-18	Classifies visual quality into High/Medium/Low
MQ-135	Provides gas-response measurements
DHT22	Records temperature and relative humidity
ESP32	Collects and communicates sensor data
Decision Layer	Combines visual and sensor information
Final Grading	Produces Grade A/Grade B/Reject
Data Storage	Records inspection information

6.3 User or stakeholder validation

Checking the system is about seeing if it works for quality control in the fish processing area.

The system can be checked by comparing its quality ratings with the ones from experts or set standards. The check should find out how much the system agrees with the expert ratings.

The check can look at:

- How easy the system is to use.
- How clear the Grade A, Grade B and Reject results are.
- How consistent the quality decisions are.
- How useful the sensor data is.
- How good the conveyor-based setup is.
- How helpful the system is for monitoring and sorting.
- How reliable the information shown to the operator is.

6.4 Integration results

Integration testing checked if the parts of hardware, AI and data processing can work together as one system.

The main process was checked from sample check to grading. The C270 webcam took the data while the MQ-135 and DHT22 collected extra sensor data. The ESP32 gathered the sensor data. The data was matched using sample ID and time.

The combined data was given to the -sensor decision part, where visual and sensor data were used to make the final quality grade.

6.5 Summary of findings

The research made an AI-based -sensor way to check the quality of dried Maldive fish as part of the bigger smart system. The method combines data, deep learning, gas response, environmental data and IoT-based data collection.

The ResNet-18 model gives the visual check by sorting dried Maldive fish into High, Medium and Low quality. The MQ-135 gives gas data while the DHT22 gives temperature and humidity. The ESP32 helps collect and send sensor data.

Combining all this data gives the base for the Grade A, Grade B and Reject ratings. The results from tests will show how good this method is.

Overall, the check should show if the system gives a more organized way to check quality than just using manual checks. The system should be seen to help with quality checks and more testing, with bigger and different data sets and real food-quality methods will be needed before using them in the market.

7 DISCUSSION

7.1 Interpretation of findings

The results show that an AI and sensor-based way can give a way to check the quality of dried Maldive fish. The system uses the C270 HD webcam for data and the MQ-135 and DHT22 sensors for extra data.

The ResNet-18 model gives the visual check by sorting the fish into High, Medium and Low quality. This shows that deep learning can help with checking parts. The model's success depends on the quality, size and variety of the image data used for training and testing.

The MQ-135 gives gas data that can be used with the visual check. The results should be seen as showing if gas data helps not as proof that the MQ-135 directly measures spoilage. Similarly, the DHT22 temperature and humidity data give the environment context that helps understand the checking conditions.

The mix of these information sources forms the base for the final Grade A, Grade B and Reject classification. This multi-sensor method is likely to give information than depending only on one source of information although the real improvement needs to be shown through the results from experiments.

7.2 Comparison with literature baselines and existing solutions

Previous studies have shown how computer vision, deep learning, sensor technologies and IoT systems can be used for food-processing and fish-quality applications. Computer-vision methods can offer automated analysis of product features while sensor-based methods can add environmental or gas-related information.

Compared to an inspection method the system offers a more organized and repeatable way to keep track of inspection details. Of only depending on what the operator sees the system makes digital records that include the sample ID, time stamp, visual classification, sensor data and final quality grade.

Compared to a camera- quality assessment system the system adds sensor information from the MQ-135 and DHT22. This lets the system look at gas-response and information along with visual features.

Compared to a sensor- approach the system still uses computer vision as the main way to assess quality. This is important because the outside look of dried fish gives useful information that sensors alone cannot provide.

However, comparing with published research should be based on the actual papers reviewed by the research group. The uploaded literature files are not available now so specific claims about individual authors, datasets, accuracy numbers or published baselines should not be made without re-uploading those sources.

7.3 Research contribution

The main research contribution is the creation of a -sensor quality evaluation approach that is specifically for dried Maldive fish.

The research contributes in ways:

1. Application of Deep Learning

The study uses a ResNet-18 model to classify the quality of dried Maldive fish into High, Medium and Low categories.

2. Multi-Sensor Quality Evaluation

The research combines computer vision with MQ-135 gas-response data and DHT22 environmental measurements.

3. IoT-Based Data Acquisition

The ESP32 offers a way to collect and share sensor information.

4. Integrated Quality Grading

The research suggests a Grade A, Grade B and Reject decision based on both visual and sensor information.

5. Digital Quality Traceability

The system provides records that include sample ID, time stamp, visual results, sensor readings and final quality grades.

6. Application to Maldive Fish Processing

The proposed method deals with the quality-evaluation needs of dried fish rather than assuming that methods for other food or fish products can be directly used.

7.4 Product contribution

From a product point of view the developed component adds an automated quality-inspection module to the AI and IoT-Based Smart Maldive Fish Processing System.

The main product contributions include:

• Automated inspection using the C270 webcam.
• AI-based quality classification using ResNet-18.
• gas monitoring using MQ-135.
• Environmental monitoring using DHT22.
• IoT-based sensor data collection using ESP32.
• Multi-sensor quality decision-making.
• Grade A, Grade B and Reject classification.
• Digital quality. Traceability.
• Potential help for automated sorting and monitoring.

The modular design also lets the system grow in the future by adding sensors, better AI models, bigger data sets or automated sorting features.

7.5 Limitations and threats to validity

Several issues could affect the validity and ability to apply the research findings.

7.5.1 Dataset Limitations

The performance of the ResNet-18 model depends on how big good. Varied the image data set is. If the data set has limited changes in fish look, position, lighting or processing conditions the model might not work the same in places.

7.5.2 Visual Classification Limitations

The camera looks at outside features. Things like problems, germs or chemical issues that you can't see through images can't be found easily just by looking at a picture.

7.5.3 MQ-135 Limitations

MQ-135 is a general gas sensor that reacts to gases. Its reading can also be changed by the environment. So, the sensor can't be used directly to measure spoilage, freshness or expiration without tests.

7.5.4 Environmental Sensor Limitations

The DHT22 measures temperature and humidity in the air. These readings don't show the moisture or full quality of the dried fish.

7.5.5 Reference-Quality Limitations

How well the Grade A, Grade B and Reject classification works depends on how good and consistent the labelers during testing. Subjective labels may add differences.

7.5.6 Integration Limitations

It can be hard to match image and sensor data when samples are moving constantly on a conveyor. Wrong sample ID or timing could make sensor data linked to the image.

7.5.7 Generalizability

Results from a controlled test might not be the same in processing places, lighting, environment, fish types or sizes of production.

7.6 Implications

The results have effects on how smart Maldive fish processing systems are developed.

7.6.1 Implications for Quality Control

The method shows how it can reduce the need for manual visual checks by offering a structured and digitally recorded way to evaluate quality. This could help make inspection more consistent.

7.6.2 Implications for Processing Industry

An automated inspection system that works on a conveyor could be used in Maldive fish processing plants to help with checks and sorting by quality. The modular setup also allows for sizes of use.

7.6.3 Implications for Data-Driven Processing

Keeping records of image classifications and sensor data makes a history of checks. Over time these records could help compare batches, track quality look at processes and develop prediction models.

7.6.4 Implications for AI Development

The study shows that combining AI-based computer vision with sensor data is possible of just using one data source. With tested data future versions could try more advanced techniques that use more than one type of data.

7.6.5 Implications for Future Research

Future work should focus on getting a bigger and more varied data set testing how MQ-135 readings relate to Maldive fish quality improving how sensors are calibrated checking environmental and gas sensors and looking at different deep learning models.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

8.1 The AI and IoT-Based Smart Maldive Fish Processing System was created to make traditional Maldive fish processing more efficient consistent monitored and better at checking quality. The whole system brings together processing, IoT-based data collection, sensor monitoring, optimization and automatic quality checks to help with a more organized way of working.

The individual part, AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish focuses on the check for quality. The method uses a C270 HD webcam, a ResNet-18 deep learning model, a MQ-135 gas sensor, a DHT22 temperature and humidity sensor and an ESP32 to get and use types of information.

The ResNet-18 model is used to split the look of dried fish into High, Medium and Low. The MQ-135 gives data about gases while the DHT22 gives info about the environment with temperature and humidity. These inputs are combined using a - sensor decision system to get the final classification of the product into Grade A, Grade B or Reject.

The system also records sample IDs, time stamps, visual classifications, sensor readings and final quality grades. This gives a base for keeping track of quality, checking, reporting and future sorting.

Overall, the research shows how using AI, computer vision, IoT and multiple sensors can help with checking quality of dried fish. However, the system should be seen as a way to check quality and help make decisions until more tests are done with more varied data sets and better ways to check quality.

8.2 Achievement of objectives

Table 8 - Achievement of Research Objectives

Objective	Achievement
Develop a computer-vision-based quality classification system	A C270 webcam and ResNet-18 approach were designed for High/Medium/Low visual classification.
Acquire supplementary gas-response information	MQ-135 was integrated to collect gas-response measurements.
Monitor environmental conditions	DHT22 was incorporated to measure temperature and relative humidity.
Develop IoT-based sensor acquisition	ESP32 was used as the sensor data-acquisition and communication device.
Integrate visual and sensor information	A multi-sensor decision layer was designed to combine the different information sources.
Develop final product grading	The system was designed to produce Grade A, Grade B, or Reject classifications.
Evaluate the proposed system	Evaluation measures including accuracy, precision, recall, F1-score, confusion matrix, sensor consistency, and grading agreement were defined.

8.3 Individual research and product contributions

he main research contribution was the creation of an AI-based -sensor method for checking quality of dried Maldive fish.

The research contribution includes using ResNet-18 for classifying quality and looking at how extra gas and environmental info can be used in the quality check process.

From a product development side, the contribution includes:

- C270 HD webcam for taking pictures.
- ResNet-18 for classifying the look of the fish.
- MQ-135 for gas data.
- DHT22 for checking temperature and humidity.
- ESP32 for getting and sending sensor data.
- Multi-sensor decision system for quality check.

• Grade A, Grade B and Reject classification.
• Digital records, for tracking and checking quality.

This part helps the group project by adding an automatic final check that can link with other parts of processing and improving.

8.4 Future directions

8.4.1 More Diverse Dataset

A bigger image dataset needs to be gathered from groups, fish sizes, angles, lighting situations and processing conditions. This could make the ResNet-18 model better at understanding situations.

8.4.2 Better Deep Learning Models

Future work can compare ResNet-18 with models and investigate transfer learning, data augmentation and multi-sensor deep learning methods to make visual and overall quality classification better.

8.4.3 Better Sensor Testing

More experiments should investigate the link between MQ-135 gas responses and measured dried Maldive fish quality. Sensor calibration and controlled experiments should happen before setting any quality limits.

8.4.4 More Sensors

Future versions could investigate sensors that give more specific details about product condition, moisture or volatile compounds. This could make the multi-sensor decision process more reliable.

8.4.5 Better Multi-Sensor Decision-Making

of just using manually created decision rules future research could look into machine-learning-based multi-sensor fusion models that learn how visual features and sensor data are related from experimentally tested datasets.

8.4.6 Automatic Sorting

The Grade A, Grade B and Reject outputs could be connected to a mechanical sorting system. This would allow the system to separate products based on their quality classification.

8.4.7 Live Monitoring Dashboard

A full dashboard could be made to show inspection results, sensor readings, batch data, quality trends and past records.

8.4.8 Link with the Whole Smart Processing System

Future work can make the connection between the quality evaluation part and the other parts of the Maldive fish processing system stronger. Processing conditions, optimization results and final quality outcomes could be combined to find out how processing parameters relate to the product quality.

8.4.9 Testing at Commercial Scale

Before putting the system into use it should be tested in actual processing environments using bigger datasets and samples from many different batches. Performance should be checked in situations to see how reliable, scalable and cost-effective it is.

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GLOSSARY

Term	Definition
Artificial Intelligence	A technology that enables computer systems to perform tasks such as analysis, classification, prediction, and decision-making.
Computer Vision	A technology that enables a computer system to obtain and analyze information from images or video.
Deep Learning	A machine learning approach that uses multi-layer neural networks to learn patterns from data.
ResNet-18	A deep learning model based on residual network architecture, used in this project to classify the visual quality of dried Maldive fish.
Convolutional Neural Network	A deep learning architecture commonly used to identify and classify patterns in images.
Image Classification	The process of assigning a captured image to a predefined category based on its visual characteristics.
Multi-Sensor System	A system that combines information collected from multiple sensors or data sources to support a decision.
Quality Evaluation	The process of assessing a product to determine whether it meets defined quality requirements.
Dried Maldive Fish	A processed fish product produced through stages such as boiling, salting, and drying.
Gas Response	The response or reading produced by a gas sensor when exposed to gases in its surrounding environment.
Environmental Monitoring	The measurement and recording of environmental conditions such as temperature and relative humidity.
Sensor Data Acquisition	The process of collecting measurements from sensors for further processing, analysis, or storage.
Data Preprocessing	The process of preparing raw image or sensor data so that it can be used for analysis or machine learning.
Multi-Sensor Decision Layer	The system stage where visual classification and sensor information are combined to determine the final quality grade.
Quality Grade	A category assigned to a product according to its evaluated quality characteristics.
Grade A	The proposed category representing dried Maldive fish classified as high quality.

Grade B	The proposed intermediate category representing dried Maldive fish with medium or uncertain quality.
Reject	The proposed category for dried Maldive fish that does not meet the defined quality requirements.
High Quality	A visual classification representing samples with acceptable high-level visual characteristics.
Medium Quality	A visual classification representing samples with intermediate visual characteristics.
Low Quality	A visual classification representing samples with comparatively poor visual characteristics.
Quality Scoring	The process of combining relevant quality information to support a final quality decision.
Conveyor-Based Inspection	An inspection method in which fish are transported through a designated inspection area using a conveyor.
Visual Quality Assessment	The evaluation of product quality based on observable characteristics obtained from captured images.
Sensor Fusion	The combination of information from multiple sensors or data sources to support a common analysis or decision.
IoT Monitoring	The use of connected devices and sensors to collect, communicate, and monitor information digitally.
Model Inference	The process of using a trained AI model to produce a prediction or classification for new input data.
Training Dataset	A collection of data used to train a machine-learning model.
Validation Dataset	A dataset used during model development to evaluate model performance and support model selection.
Test Dataset	A separate dataset used to evaluate the performance of a trained model on previously unseen data.
Confusion Matrix	A table used to show correct and incorrect predictions for different classification categories.
Data Synchronization	The process of ensuring that image and sensor information are correctly associated with the same sample or inspection event.
Traceability	The ability to track an inspection result back to the relevant sample and recorded information.
Quality Control	The systematic process of monitoring and evaluating products to ensure that they meet defined quality requirements.

APPENDICES

Appendix A Research instruments or extended results

This appendix contains supplementary materials related to the development, implementation, and testing of the AI-Based Multi-Sensor Quality Evaluation System for Dried Maldive Fish. The materials provide supporting evidence for the methodology and results presented in the main report.

Hardware and Research Instruments

Instrument	Purpose
C270 HD Webcam	Capturing images of dried Maldive fish for visual quality classification
MQ-135 Gas Sensor	Collecting supplementary gas-response information
DHT22 Sensor	Measuring temperature and relative humidity
ESP32	Sensor data acquisition and communication
Conveyor Belt	Moving dried Maldive fish through the inspection area
Computer/Processing System	Processing images and running the ResNet-18 model

Appendix B Academic integrity and AI use disclosure

B.1 AI Tools Used

Tool: OpenAI ChatGPT

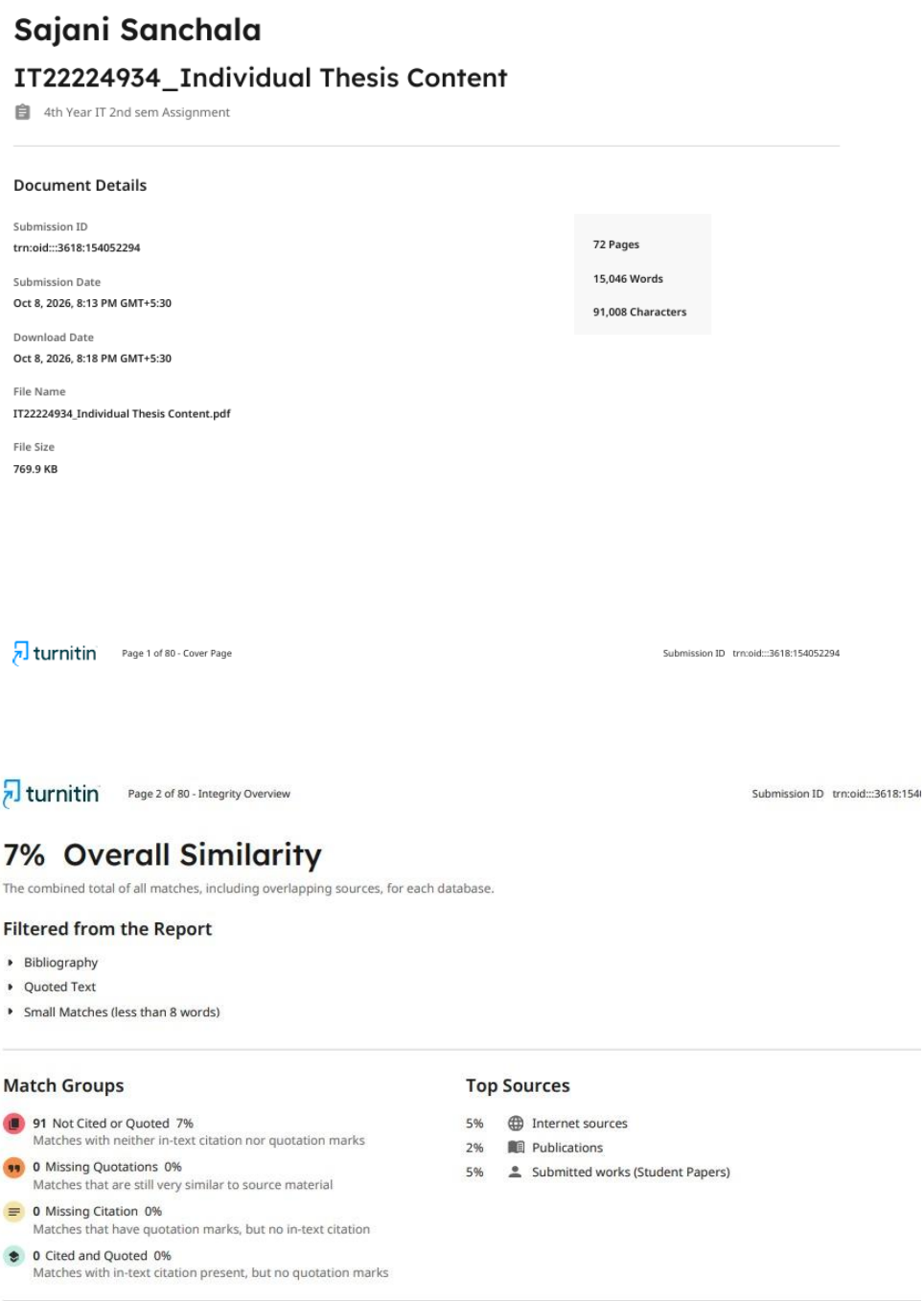
ChatGPT was used as a supporting tool during the preparation of the research report.

B.2 Purpose of AI Use

The tool was used for:

- Improving grammar and academic wording.
- Structuring report sections.
- Refining explanations of technical concepts.
- Organizing methodology and system descriptions.

- Developing preliminary descriptions of diagrams and workflows.
- Assisting with table and section formatting.
- Reviewing and improving draft content.



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B.3 Verification

The student is responsible for checking the accuracy of all final content. In particular:

- Actual experimental results must be obtained from the project testing.
- Numerical performance values must not be fabricated.
- Hardware and software descriptions must match the implemented system.
- References must be checked against the original sources.
- Sensor capabilities and limitations must be accurately represented.
- MQ-135 readings must not be presented as universal spoilage or expiry thresholds without experimental validation.
- AI-generated explanations must be reviewed before inclusion in the final report.

B.4 Sections and Artefacts Affected

AI assistance was used during the preparation and refinement of several parts of the report, including:

- Introduction and background
- Literature review
- Research methodology
- Product and solution design
- Implementation and testing descriptions
- Results and validation structure
- Discussion and conclusion
- Glossary
- Appendix wording
- Diagram and table descriptions